

## Problem Set for 08 May 2025: A New Jordan form over $\mathbb{C}$

### Important

Throughout,  $\mathbb{C}$  is the base field. Define  $|z| := \sqrt{z\bar{z}}$  for  $z \in \mathbb{C}$ .

**Definition** (Eigen-). Let  $A \in \mathbb{C}^{n \times n}$ . We say that  $(v, \lambda)$  is a eigenpair of  $A$  provided  $A\bar{v} = \lambda v$  for some  $\lambda \in \mathbb{C}$ . The vector  $v$  is referred to as a eigenvector and  $\lambda$  a eigenvalue, correspondingly. For matrices, we say  $A$  is csimilar to  $B$  provided  $A = P\bar{B}P^{-1}$  for some non-singular matrix  $P$ .

### Note

You can firstly try Exercise 13-15.

**Exercise 1** Characterise the eigenspace corresponding to the eigenvalue 0.

### Warning

Eigenspaces associated with eigenvalues  $\neq 0$  are not, in general,  $\mathbb{C}$ -vector spaces.

**Exercise 2** Demonstrate that if  $\lambda$  is a eigenvalue, then so too is  $e^{i\theta} \cdot \lambda$ . Furthermore, show that for any eigenvector  $v$ , there exists a eigenvector associated with a eigenvalue  $\geq 0$ .

**Exercise 3** Suppose  $(v, \sigma)$  is a eigenpair of  $A$ ; then  $(v, \sigma\bar{\sigma})$  is an eigenpair of  $A\bar{A}$ . In particular,

- if  $A \in \mathbb{C}^{d \times d}$  possesses  $d$  eigenpairs which are linearly independent over  $\mathbb{C}$ , then  $A\bar{A}$  is diagonalisable with all eigenvalues non-negative.

Now consider the following **partial converse statement**:

- $A$  has no eigenvectors if and only if  $A\bar{A}$  admits no eigenvalues in  $\mathbb{R}_{\geq 0}$ ;
- $A \in \mathbb{C}^{d \times d}$  has  $d$  linearly independent eigenpairs over  $\mathbb{C}$  if and only if  $A\bar{A}$  is diagonalisable with non-negative eigenvalues.

Prove the **partial converse statement**: suppose that  $(v, \sigma\bar{\sigma})$  is an eigenpair of  $A\bar{A}$ ; then  $\sigma$  is a eigenvalue of  $A$ .

### Tip

Construct an associated eigenvector of some  $e^{i\theta}\sigma$ , then invoke previous exercises.

### Warning

When  $A$  has geometric multiplicity 1, one may simply take the eigenvector  $e^{i\theta}v$ ; however, in general,  $e^{i\theta}v$  is not a eigenvector. The construction closely resembles that found in **Exercise 6**.

**Exercise 4** Elucidate **Exercise 3** (particularly the warning) with the following example:

$$A = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

**Exercise 5** This serves to complete **Exercise 2**. Suppose  $\{(v_i, \lambda_i)\}_{i=1}^n$  are eigenpairs of  $A$  such that  $|\lambda_i| \neq |\lambda_j|$ ; show that  $\{v_i\}_{i=1}^n$  are linearly independent.

**Exercise 6** We aim to identify the analogue of the Jordan canonical form in the c-version, namely,  $X = P\bar{J}P^{-1}$ . We commence with a lemma:

- $P\bar{P}^{-1} = I$  if and only if  $P = Q\bar{Q}^{-1}$  for some invertible  $Q$ .

 **Tip**

Take  $Q = a\bar{P} + bI$ .

**Exercise 7** The most elementary Jordan matrix is the diagonal matrix. Show that  $A$  is cdiagonalisable (i.e.,  $A = P D \bar{P}^{-1}$  for some  $P$ ) if and only if  $A\bar{A}$  is diagonalisable. In this scenario,  $A\bar{A}$  has only non-negative eigenvalues.

 **Note**

In this diagonalisable case, one may also verify that  $\text{rank}(A) = \text{rank}(A\bar{A})$ ; that is, the multiplicity of 0 is preserved.

**Exercise 8** (Nilpotent part). Show that there exists a matrix  $P$  such that  $P A \bar{P}^{-1} = \begin{pmatrix} N & O \\ O & B \end{pmatrix}$ , where  $N$  comprises nilpotent Jordan blocks and  $B$  is invertible.

 **Tip**

Consider the method we used last term to find  $J_{s_i}(0)$ 's. Useful hints can be found in **Exercise 11**.

**Exercise 9** To analyse the Jordan form of  $A$ , we begin by studying the Jordan form of  $A\bar{A}$ . Demonstrate the following:

1.  $A\bar{A}$  has a characteristic polynomial with real coefficients;
2. non-real Jordan blocks occur in conjugate pairs;
3. if  $\dim \ker(\lambda I - A\bar{A}) = 1$  and  $\lambda \in \mathbb{R}$ , then  $\lambda \geq 0$ ;
4. **(optional)** Generalise item 3 by showing that for any eigenvalue  $\lambda < 0$ , the dimension of the null space  $\dim \ker(\lambda I - A\bar{A})^p$  is always even ( $\forall p \geq 1$ ), implying that negative Jordan blocks occur in pairs.

 **Warning**

This is a straightforward exercise. Nonetheless, note that  $A\bar{A}$  may possess negative eigenvalues.

**Exercise 10** Show that there exists a matrix  $P$  such that  $P A \bar{P}^{-1}$  is block diagonal with the following summands:

1. (Jordan type):  $J_s(\lambda)$  where  $\lambda \geq 0$ ;
2. (Half-Jordan type): Let  $H_{2s}(\lambda) := \begin{pmatrix} O & J_s(\lambda) \\ I & O \end{pmatrix}$ .
  1. (For non-real eigenvalues): Since each complex Jordan block in  $A\bar{A}$  appears in a pair, i.e.,  $\begin{pmatrix} J_s(z) & O \\ O & J_s(\bar{z}) \end{pmatrix}$ , define  $H_{2s}(z)$  as a block of  $P A \bar{P}^{-1}$  satisfying

$$H_{2s}(z) = \begin{pmatrix} O & I \\ J_s(z) & O \end{pmatrix}.$$

2. (For negative eigenvalues): For a negative eigenvalue  $\lambda$  of  $A$ , the Jordan blocks also appear in pairs (see **Exercise 9**).

We refer to this as the cJordan form of  $A$ . It is uniquely determined by  $A$ .

 **Tip**

The uniqueness (as we equal  $H(z)$  and  $H(\bar{z})$ ) of non-zero  $J$ - and  $H$ -blocks is governed by  $A\bar{A}$ ; the uniqueness of the  $J(0)$ -portion is determined by the rank sequence  $\text{rank}(A)$ ,  $\text{rank}(A\bar{A})$ ,  $\text{rank}(A\bar{A}A)$ , and so forth.

**Exercise 11 (optional)** (The uniqueness between the cJordan form of  $A$  and the Jordan form of  $A\bar{A}$ ). The cJordan form is essentially unique.

- Find distinct matrices  $A \neq B$  such that  $A\bar{A} = B\bar{B}$ ;
- Suppose  $A\bar{A} = B\bar{B}$  are invertible; then there exists a matrix  $P$  such that  $P A \bar{P}^{-1} = B$ . In this case, we say  $A$  is scimilar to  $B$ .

In summary,  $A$  is csimilar to  $B$  if and only if the following hold:

1.  $A\bar{A}$  is similar to  $B\bar{B}$ ;
2.  $\text{rank}(A\bar{A}A \cdots A(\bar{A})) \equiv \text{rank}(B\bar{B}B \cdots B(\bar{B}))$ ; that is,  $A$  and  $B$  possess the same  $J(0)$ -portion in their cJordan forms.

**Exercise 12 (optional)** Utilise the cJordan form to show that any complex matrix  $A$  is csimilar to:

1.  $\lambda A$  for  $|\lambda| = 1$ ;
2.  $\bar{A}$ ;
3.  $A^T$ ;
4. a real matrix;
5. a Hermitian matrix.

**Exercise 13 (optional)** Show that for any complex square matrix  $A$ ,

$$\det(AA^H + I) \geq \det(A\bar{A} + I) > 0.$$

 **Tip**

Apply the identity  $AA^T + BB^T = RR^T$ , and the Cauchy–Binet formula to establish the first inequality.

**Exercise 14 (optional)** Using cJordan forms, show that any complex matrix is the product of a symmetric matrix and a Hermitian matrix.

 **Note**

It is known that any real or complex matrix can be expressed as the product of two symmetric matrices (see exercises from the past semesters).

**Exercise 15 (optional)** Show that  $\begin{pmatrix} O & A \\ \bar{A} & O \end{pmatrix}$  is similar to  $\begin{pmatrix} O & B \\ \bar{B} & O \end{pmatrix}$ , if and only if  $\begin{pmatrix} O & A \\ -\bar{A} & O \end{pmatrix}$  is similar to  $\begin{pmatrix} O & B \\ -\bar{B} & O \end{pmatrix}$ .

 **Tip**

Equivalently, if and only if  $A$  and  $B$  are csimilar.